Solangi Mohanty

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.14
FIITJEE Inter College	Class XII (Telangana State Board)	2023	94.30%
FIITJEE World School	Class X (Telangana State Board)	2021	10.00%

PROFESSIONAL EXPERIENCE

Machine Learning Engineering Intern | ServiceNow

- Engineered and contributed production code to a Python and FastAPI-based multi-agent orchestrator service.
- Enabled dynamic routing of user queries to appropriate external tools and Large Language Models, including GPT-4o, Gemini, and Claude.
- Developed an end-to-end evaluation dataset and authored Python automation scripts to benchmark AI response quality, replacing manual QA with a scalable, repeatable testing pipeline.
- Debugged and resolved failing automated integration tests across the request pipeline, stabilizing the test suite to ensure reliable code merges.
- Utilized Server-Sent Events (SSE) streaming over REST APIs to deliver real-time chunked responses back to chat clients.
- **Tech Stack:** Python, FastAPI, Docker, Git/GitHub, REST APIs, Server-Sent Events (SSE), LLMs (GPT-4o, Gemini, Claude).

SKILLS

Programming Languages: C/C++, Python, SQL

Databases: MySQL

Development Tools: Git/Github, Visual Studio Code, Anaconda, Jupyter Notebook

Libraries: NumPy, Pandas, Matplotlib, Scikit-learn, FFmpeg, XGBoost

PROJECTS

Online Payment Fraud Detection

- Used a dataset containing information such as account balance of sender and receiver before and after a transaction
- Predicted fraudulent transactions using XGBoost, RandomForestClassifier, SVC and LogisticRegression
- Evaluated performance by testing the models on training and validation sets and visualising using confusion matrix

Buffon's Matches -- Monte Carlo Estimation

- Generated synthetic throws (random center & angle) on parallel lines and detected crossings via nearest-line distance vs projected half-length.
- Built an animated visualisation: colour-coded crossed (red) vs not crossed (green), showed live counts (throws, crossings) and the running estimate; saved the simulation as MP4.
- Analysed convergence/error vs. sample size; exported runs to CSV/GIF for reporting.

Birthday Paradox Probability Estimator & Visualiser

- Implemented a Python/NumPy program to compute the famous Birthday Problem**.**
- Compared the exact curve to the analytic **Poisson approximation** and explained the similarity.

- Addressed numerical precision with stable iterative products and sanity checks to avoid floating-point drift on larger n.

SCHOLASTIC ACHIEVEMENTS

Qualified KVPY Examination with **AIR 1948**

State Topper in BrainOBrain Wonderkid, 2018

Received **Certificate of High Distinction** in International Benchmark Test for the subjects of Mathematics, Reasoning, Science and English

Qualified Unified Cyber Olympiad, 2017 with **AIR 32**

Secured a **National Rank of 32** in WIZ National Spell Bee, 2017

RELEVANT COURSES

Computer Science: Introduction to Programming, Introduction to Artificial Intelligence, Statistics, Probability and Random Variables, Data Structure and Algorithms, Object Oriented Programming, DBMS, Operating Systems, Theory of Computation

Mathematics: Calculus, Differential Equations, Maths for Physics, Linear Algebra, Complex Variables

POSITIONS OF RESPONSIBILITY

Outreach & Collaborations Head at TEDxIITHyderabad

Partnership & Collaboration (Sponsorship) Coordinator at TedxIITHyderabad

Operations Manager at Ecell, IIT Hyderabad

Interview & Hosting Coordinator at Media Council, Gymkhana

EXTRACURRICULARS

Inter IIT Team Member in **Badminton** for **IISM, 2025** at IIT Madras and **IISM, 2024** at IIT Indore

Inter IIT Team Member in the **Dance Arts** (Group Dance & Street Battle) for the **IICM, 2024** at IIT Patna.

Inter IIT Team Member in the **Dance Arts** (Group Dance and Street Battle) for the **IICM, 2024** at IIT Patna.

Inter IIT Team Member in the **Dance Arts** (Group Dance) for the **IICM, 2023** at IIT Kharagpur.

2nd Prize in Duo Dance, Elan & nVision (College fest of IIT Hyderabad), 2024.

3rd Prize in Group Dance, Elan & nVision (college fest of IIT Hyderabad), 2025.

3rd Prize in Group Dance, Battle of Beats 2025, a dance competition in Mahindra University.